

2(a)(i)	continuous ; A normal distribution	1
2(a)(ii)	evolution / natural selection / artificial selection / directional selection / adaptation / selective breeding ;	1
2(b)	any three from: <i>captive fish:</i> 1 controlled conditions so phenotype less varied ; 2 more oxygen so larger ; 3 less space / overcrowding, so smaller ; 4 less space / overcrowding, so (signs of) disease / wounds ; 5 food qualified linked to, size / muscle / fat ; 6 AVP ; e.g. named tank condition linked to skin colour / no predators linked to increase in size / temperature linked to size	3

Question	Answer	Marks
2(c)(i)	any five from: 1 obtain mRNA from wild and captive bred fish / AW ; 2 reverse transcription of mRNA to produce <u>c</u>DNA ; 3 add fluorescent label to (c)DNA ; I colour A dyes / markers / tags 4 (microarray has) ssDNA <u>probes</u> ; 5 each from a, different / known, gene ; 6 (c)DNA hybridises / AW, to, probes / ssDNA (on microarray) ; 7 fluorescence shows the expressed genes ; 8 intensity of fluorescence (shows level of gene expression) ; 9 compare, fluorescence / gene expression (between fish groups) ; 10 AVP ; e.g. <i>ref. to</i> washing off excess <u>(c)DNA</u> (only), after hybridisation UV light / laser scanning (to record fluorescence pattern)	5

Question	Answer	Marks
2(c)(ii)	1 <i>ref. to</i> transcription factors ; <i>any three from:</i> 2 TF / gene product / protein, bind to DNA ; 3 (TF binds to) promoter ; 4 switches genes, on / off ; AW I genes, expressed / not expressed 5 <i>ref. to</i> binding of RNA polymerase ; R if operator mentioned 6 mRNA is made / transcription occurs ; ora 7 AVP ; e.g.(so that genes are expressed) at the correct time / in the correct context / in the correct cell type / in the correct order ;	4
2(d)(i)	<i>any one from:</i> 1 differences caused by selection in captive-bred population ; 2 (not environmental because) all, offspring / fish, were kept in same environment ;	1
2d(ii)	<i>idea that</i> captive fish have better, wound healing / immune response ;	1